

Supplementary Material C: Sensitivity Analysis by Quality Level

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Overview

To assess whether the review’s conclusions depend on study quality, we stratified the 83 included studies by their total quality appraisal score and re-examined the prevalence of each responsible AI dimension. Quality scores ranged from 4.0 to 5.0 (out of 5.0), with all studies falling into the medium or high categories.

Quality Score Distribution

Total QA Score	Number of Studies	Proportion
5.0	14	16.7%
4.5	37	44.0%
4.0	32	38.6%

Table 1: Distribution of total quality appraisal scores across the 83 included studies.

Responsible AI Dimensions by Quality Level

Table 2 reports the percentage of studies addressing each responsible AI dimension, stratified by quality score.

Quality Score	Data Quality	Seq. Stability	Interpretability	All Three
5.0 (n=14)	35.7%	78.6%	50.0%	0.0%
4.5 (n=37)	37.8%	81.1%	32.4%	2.7%
4.0 (n=32)	56.2%	90.6%	31.2%	9.4%
All (n=83)	44.6%	84.3%	34.9%	4.8%

Table 2: Percentage of studies addressing each responsible AI dimension, stratified by total quality appraisal score.

Responsible AI Dimensions by Quality Threshold

Table 3 presents a cumulative threshold analysis, showing results when restricting the corpus to studies at or above a given quality score.

Threshold	Data Quality	Seq. Stability	Interpretability	All Three
≥ 4.0 (n=83, all studies)	44.6%	84.3%	34.9%	4.8%
≥ 4.5 (n=51)	37.3%	80.4%	37.3%	2.0%
$= 5.0$ (n=14)	35.7%	78.6%	50.0%	0.0%

Table 3: Percentage of studies addressing each responsible AI dimension when restricting to studies at or above a quality threshold.

Interpretation

The core findings of the review remain consistent across quality levels:

1. **Sequential stability** is addressed by a majority of studies regardless of quality level (78.6%–90.9%), confirming that this dimension has achieved broad methodological acceptance across the field.
2. **Data quality** consideration varies across quality levels (35.7%–57.6%), but in all strata a substantial proportion of studies neglect data quality concerns. The higher attention to data quality among lower-scoring studies may reflect that these studies discuss data issues without the methodological depth required for higher quality scores on other criteria.
3. **Interpretability** remains a minority concern across all quality levels (30.3%–50.0%), supporting the conclusion that the field has not broadly adopted process-level interpretability mechanisms.
4. **Simultaneous attention to all three dimensions** is rare across all strata (0.0%–9.1%), confirming the fragmented approach to responsible AI observed in the full corpus.

These results indicate that the review’s principal conclusions—widespread neglect of data quality, low adoption of interpretability, and fragmented approaches to responsible AI—are robust to variation in study quality and are not driven by a particular quality stratum.